REQUIREMENT GATHERING

Date: 19/July/2026

1. **Project Overview**:
   LittleLumin is an AI-powered child development, parenting support, and digital therapeutic platform designed to promote healthy cognitive, emotional, behavioral, and developmental growth in children aged 3-6. Built on principles of developmental psychology, occupational therapy, and adaptive learning, the platform provides personalized experiences for children while empowering parents, teachers, therapists, and child specialists with meaningful insights and evidence-based recommendations.

   Problem Statement:
   Despite the abundance of educational apps for young children, there is a critical gap in platforms that address the dual needs of both child development and parenting competence. Existing solutions either place children directly on screens (causing health and developmental concerns) or provide static, one-size-fits-all content that ignores individual learning patterns. Furthermore, parents—especially first-time Gen Z parents—lack the knowledge, confidence, and daily guidance needed to actively participate in their child's developmental journey.

   Main Objectives:
   - To generate personalized, screen-free therapeutic activities in real-time based on each child's unique learning pattern
   - To track child development across multiple domains using VABS-II clinical assessment
   - To empower parents with emotional wellbeing support and growth tracking
   - To provide professionals (OTs, psychologists) with remote monitoring and recommendation capabilities
   - To support both typically developing children and those needing additional therapeutic intervention

2. **System Scope**:

The LittleLumin platform is proposed as a full-scale implementation with a phased development approach:

Phase Scope

- Mini Project : Working mobile prototype with 5 core AI modules and 3 user roles (Parent, Child, Admin)
- Main Project : Complete production-ready platform with 14 AI modules, 7 user roles, mobile + web dashboards

The platform will serve as:

- A research-backed therapeutic tool
- A commercial parenting guidance platform
- A professional support system for therapists and educators

3. **Target Audience**:

Category	**Description**
Parents	Primary caregivers of children aged 3-6, including working parents and stay-at-home parents
Children	Children aged 3-6 (guided offline by parents, zero screen time)
Teachers	Preschool and kindergarten educators
Occupational Therapists	Professionals supporting child development

Child Psychologists	Mental health professionals working with children
Primary Care Providers	Healthcare professionals supporting caregiver wellbeing
Administrators	Platform managers and content moderators
Content Creators	Educators and therapists creating activity content
Researchers	Academic and clinical researchers studying child development

4. **Modules**:

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	Module	**Functionality**
Core Adaptive Engine		
1	Procedural Therapeutic & Enrichment Activity Generator	Generates personalized activities in real-time using AI
2	Adaptive Difficulty & Performance Analyzer	Predicts difficulty, analyzes feedback, updates skill profile
3	Growth Pathway & Progress Forecaster	Identifies growth areas, forecasts future development

4	Milestone & Activity Recommender	Compares progress against milestones, recommends activities
Writing & Motor Skills		
5	Multi-Modal Writing Engine with Image Analysis	Generates writing activities (sand, fabric, chalk, paint, pencil) + handwriting image analysis
6	Attention & Command Tracker	Tracks attention span, command-following capacity
7	Reward Sensitivity Analyzer	Identifies motivation patterns
Content & Engagement		
8	Natural Language Story Generator	Generates custom stories, rhymes, and therapeutic narratives
9	Emotion & Engagement Analyzer	Analyzes child engagement patterns
10	Content Management & Curation Engine	Manages activity library, approves submissions
Professional & Caregiver		
11	Occupational Therapist Dashboard	Remote monitoring, therapeutic oversight, alerts
12	Psychologist / Primary Care Provider Dashboard	Holistic child + caregiver support
13	Caregiver Wellbeing & Growth Tracker	Tracks parental confidence, emotional resilience

14	Early Intervention Identifier	Flags children who may benefit from professional evaluation

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5. **User Roles**:

#	Role	Platform	Access Level
1	**Parent**	Mobile + Web	View child progress, guide activities, submit feedback, access wellbeing support
2	**Child**	Offline (guided)	Participates in parent-guided activities (zero screen time)
3	**Teacher**	Web Dashboard	Monitor classroom progress, identify students needing support
4	**Occupational Therapist**	Web Dashboard	Remote patient monitoring, therapeutic activity oversight, developmental alerts
5	**Psychologist / Primary Care Provider**	Web Dashboard	Holistic child + caregiver mental health support, recommendations
6	**Administrator**	Web Dashboard	User management, content moderation, platform analytics
7	**Content Creator**	Web Dashboard	Create activity templates, manage curriculum

6. **System Ownership**:
   - Owner : Arun Biju Thomas

7. **Industry/Domain**:

Primary Domain: Educational Technology (EdTech) / Digital Therapeutics

Secondary Domains:

- Child Development Psychology
- Occupational Therapy
- Artificial Intelligence in Healthcare
- Parenting Guidance Technology

9. **Data Collection Contacts**:

Name	**Role**	**Contact**
Ms. Mintu Mini Mathew	Speech Therapist	9562542033
Ms. Alita Mariam Varghese	Clinical Psychology Researcher	9072321768
Mr. Albin Joseph	Young Adult	9446849902
Ms. Ann Rose	Young Ault	8281254262
Ms. Alphimol Devasia	Parent	9447836441
Ms. Minnu Mini	Occupational Therappist	7306419036

10. **Questionnaire for Data Collection**:

Category 1: Working Parents

1. What is your age, and how many children do you have?

2. What is the single biggest challenge you face balancing work and parenting?

3. Would you use an app that gives you ready-made, personalized activities to do with your child in just 10–15 minutes?

4. What is the ONE feature such an app must have for you to use it daily?

Category 2: Stay-at-Home Parents

1. What is your age, and how many children do you have?

2. What is the most challenging part of being a stay-at-home parent?

3. Would you use an app that suggests new, personalized activities to do with your child every day?

4. What would stop you from using a parenting app like this?

Category 3: Expecting Parents

1. What is your age, and is this your first pregnancy?

2. What scares you most about becoming a parent?

3. Would you use an app that guides you on age-appropriate activities for your child from birth?

4. What is the ONE thing you wish such an app would do for you?

Category 4: Occupational Therapists & Child Specialists

1. What is your profession, and how many years of experience do you have?

2. What is the most common concern parents bring to you about their child's development?

3. Would you recommend a platform that generates personalized, therapeutic-style activities for parents to do at home?

4. What is the ONE feature such a platform must have to be clinically useful for you?

Category 5: Future Parents (Young Adults)

1. What age group do you belong to?

2. What part of becoming a parent feels most unknown or heavy to you right now?

3. Would you want a quiet, non-judgmental guide to help you through those moments?

4. What is one thing you wish such a guide would teach you before you become a parent

GEOTAGGED PHOTOS



